# Credit card fraud detection using machine learning

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## A. ABSTRACT

Credit card fraud detection is a critical issue for financial institutions, with fraud incidents increasing in complexity and volume as digital payment systems expand. Traditional rule-based detection methods, while useful, often fail to keep up with sophisticated fraud techniques, resulting in high false-positive rates or missed fraud cases. To overcome these limitations, machine learning (ML) algorithms and deep learning models are now widely applied to create adaptive, data-driven fraud detection systems. Here we evaluate some of the Machine learning techniques including linear regression, decision trees, random forests, and we use Deep learning models such as deep neural networks (DNN) for their effectiveness in detecting fraudulent transactions within credit card dataset.

We use Linear regression is a machine learning algorithm due to its simplicity, but its inability to capture complex patterns limits its effectiveness in this domain. We use more advanced ML algorithms, such as decision trees and random forests, are explored. Decision trees offer interpretable, rule-based classification, while random forests, an ensemble of multiple decision trees, improve detection accuracy by reducing over fitting and increasing robustness. Further we use deep learning models such as a deep neural network (DNN) for its ability to model non-linear relationships and learn hierarchical representations, which makes it well-suited for identifying the transaction anomalies indicative of fraud.

For model training and evaluation, we use a labeled dataset which is taken from kaggle containing historical transaction records with 492 fraudulent and non-fraudulent. The models are assessed using performance metrics, including accuracy, precision, recall, and F1 score, to determine their ability to effectively differentiate between legitimate and fraudulent transactions. Results indicate that while traditional ML models such as random forests and decision trees show strong performance in detecting fraud, deep neural networks achieve higher accuracy in identifying complex fraud patterns with fewer false positives. By comparing these metrics, we identify the most effective algorithm for maximizing fraud detection accuracy and minimizing false positives within the given dataset.

## B. INTRODUCTION

Credit card fraud is a significant problem, with billions of dollars lost each year.Machine learning can be used to detect credit card fraud by identifying patterns that are indicative of fraudulent transactions. Credit card fraud refers to the physical loss of a credit card or the loss of sensitive credit card information.Many machine learning algorithms can be used for detection. This project proposes to develop a machine-learning model to detect credit card fraud. The model will be trained on a dataset of historical credit card transactions and evaluated on a holdout dataset of unseen transactions.

We train this project on the dataset contains transactions made by credit cards in September 2013 by European cardholders.This dataset presents transactions that occurred in two days, where we have 492 frauds out of 284,807 transactions. The dataset is highly unbalanced, the positive class (frauds) account for 0.172% of all transactions. It contains only numerical input variables which are the result of a PCA transformation. Unfortunately, due to confidentiality issues, we cannot provide the original features and more background information about the data. Features V1, V2, ... V28 are the principal components obtained with PCA, the only features which have not been transformed with PCA are 'Time' and 'Amount'. Feature 'Time' contains the seconds elapsed between each transaction and the first transaction in the dataset. The feature 'Amount' is the transaction Amount, this feature can be used for example dependant cost-sensitive learning. Feature 'Class' is the response variable and it takes value 1 in case of fraud and 0 otherwise. We use histograms to visualize the data.

Since the data contains fraud and it is highly imbalance we use data pre processing technique to balance the imbalanced data and then we split the data as train and test data to implement the machine learning algorithms such as Linear regression, Random forest classifier and decision trees and apart from these we use Deep learning algorithms such as Deep neural networks these algorithms are used for finding the fraud transactions on the dataset.

The results are then evaluated based on performance metrics, such as accuracy, precision, recall, and F1 score, to determine their ability to effectively differentiate between non fraudulent and fraudulent transactions. We also use confusion matrix for the prediction of each machine learning and deep learning model.

## C. BACKGROUND OR RELATED WORK

We need to have some basic knowledge about the machine learning and deep learning algorithms before implementing the project. Machine learning algorithms implemented in the project are Linear regression, Random forest classifier and Decision trees and deep learning based algorithms are deep neural networks.

**Machine learning** is a subset of artificial intelligence

that involves the development of algorithms that enable computers to learn from and make predictions or decisions based on data. In the context of credit card fraud detection, ML has become an essential tool for financial institutions seeking to minimize losses due to fraudulent transactions and to enhance the security of their payment systems. The dynamic nature of fraud tactics, along with the vast volume of transaction data generated daily, necessitates the use of advanced ML techniques for effective fraud detection. we use various machine learning algorithms such as

**Linear regression** is one of the simplest and most commonly used statistical methods for predicting a dependent variable based on one or more independent variables. In the context of credit card fraud detection, linear regression can serve as a baseline model to understand the relationship between transaction characteristics and the likelihood of fraud. Despite its simplicity, linear regression has limitations in its application to fraud detection due to the complex and non-linear nature of the data involved.

**Random Forest algorithm** is a powerful tree learning technique in Machine Learning. It works by creating a number of Decision Trees during the training phase. Each tree is constructed using a random subset of the data set to measure a random subset of features in each partition. This randomness introduces variability among individual trees, reducing the risk of overfitting and improving overall prediction performance.

Random Forest is a ml algorithm used for credit card fraud detection techniques. Its ability to handle large datasets, manage class imbalance, and capture complex relationships among features makes it particularly effective in identifying fraudulent transactions. By leveraging the strengths of multiple decision trees, Random Forest not only enhances predictive accuracy but also provides insights into the factors driving fraud, thereby aiding financial institutions in safeguarding against fraudulent activities .

**Decision Trees** are a popular machine learning algorithm used for classification and regression tasks. They model decisions and their possible consequences as a tree-like structure, making them intuitive and easy to interpret. In the context of credit card fraud detection, decision trees can effectively classify transactions as fraudulent or legitimate based on various transaction attributes. We use Decision Trees in credit card fraud detection due to their interpretability, flexibility, and ability to handle various data types. While they serve as a foundational method in machine learning, their limitations necessitate careful tuning and validation. In many cases, Decision Trees are used in conjunction with other algorithms, such as ensemble methods, to enhance performance and overcome their inherent challenges. By leveraging Decision Trees, financial institutions can develop more effective fraud detection systems that are both efficient and understandable, ultimately helping to safeguard against fraudulent activities.

**Deep learning** a subset of machine learning, involves the use of neural networks with multiple layers (hence "deep") to model complex patterns and representations in data. It has emerged as a powerful tool for various applications, including credit card fraud detection, where traditional machine learning methods may struggle to capture intricate relationships within large and high-dimensional datasets. we use various deep learning algorithms

**Deep Neural Networks** (DNNs) are a class of deep learning models characterized by their multiple layers of interconnected neurons. These architectures are particularly effective for complex pattern recognition tasks, such as credit card fraud detection, where distinguishing between legitimate and fraudulent transactions requires analyzing large amounts of data and identifying subtle anomalies. Deep Neural Networks uses aggressive approach to credit card fraud detection, using their ability to model complex patterns and relationships within large datasets. By integrating DNN into fraud detection systems, financial institutions can enhance their ability to identify fraudulent transactions, improve operational efficiency, and maintain consumer trust in digital payment systems.

## D. APPROACH

This section gives a structured approach for implementing a credit card fraud detection system utilizing machine learning. The process involves several key steps, including loading the dataset, pre processing the data, splitting the data into training and testing sets, training various models, and evaluating the results to determine the effectiveness of each model.

### D.1. Load Dataset:

The first step involves loading the dataset, which contains historical credit card transaction records. In this case, the dataset can be obtained from an open-source repository such as Kaggle. The dataset typically includes features related to the transactions, such as transaction amount, time, and various transformed attributes. For now the dataset is in the commonly used European credit card fraud dataset, there are 31 attributes representing different aspects of transactions and a binary "Class" attribute indicating whether a transaction is fraudulent (1) or non fraudulent (0).

### D.2. Data Preprocessing:

Data preprocessing is a critical step in preparing the data for model training. This stage may include Balancing the Dataset Given that fraudulent transactions are often rare compared to legitimate ones, consider applying techniques such as oversampling the minority class (fraudulent trans-

actions) or under sampling the majority class (legitimate transactions) to address class imbalance.

### D.3. Split the Data:

Once preprocessing is complete, the next step is to split the dataset into training and testing sets. This allows for model training on one subset of data and evaluation on another to assess generalization performance. A common split is 70% for training and 30% for testing.

### D.4. Train Models :

Next, various machine learning models are trained on the training dataset. Common algorithms for credit card fraud detection include:

- Logistic Regression
- Decision Trees
- Random Forests
- Deep Neural Networks (DNN)

### D.5. Evaluating the results:

We use histograms to represent the features of the dataset and after training the models, it is crucial to evaluate their performance using the testing dataset. Common evaluation metrics for fraud detection include accuracy, precision, recall, F1 score, and the area under the Receiver Operating Characteristic (ROC-AUC) curve and also heat map of confusion matrix of each model for both test data and whole dataset.

## E. CODE REUSE

We have used code from the dimtics/Fraud-Detection-in-Credit-Card-Transactions-using-Machine-Learning-Techniques , for just initial part of the experiment that is the data describing and we used the logistic regression and random forest as an idea for our experiment and then we https://www.geeksforgeeks.org for identifying how to implement the models. Then we used smote over sampling technique to balance the highly imbalanced data and then we implemented the both machine learning and deep learning algorithms.

## F. EXPERIMENT

For the experiment first we took a credit card transaction dataset from the open source website https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud/data. We train this project on the dataset contains transactions made by credit cards in September 2013 by European cardholders.This dataset presents transactions that occurred in two days, where we have 492 frauds out of 284,807 transactions. The dataset is highly unbalanced, the positive class (frauds) account for 0.172% of all transactions. It contains only numerical input variables which are the result of a PCA transformation. Unfortunately, due to confidentiality issues, we cannot provide the original features and more background information about the data. Features V1, V2, … V28 are the principal components obtained with PCA, the only features which have not been transformed with PCA are 'Time' and 'Amount'. Feature 'Time' contains the seconds elapsed between each transaction and the first transaction in the dataset. The feature 'Amount' is the transaction Amount, this feature can be used for example dependant cost-sensitive learning. Feature 'Class' is the response variable and it takes value 1 in case of fraud and 0 otherwise.

We used histograms to visualize all these features of the dataset.

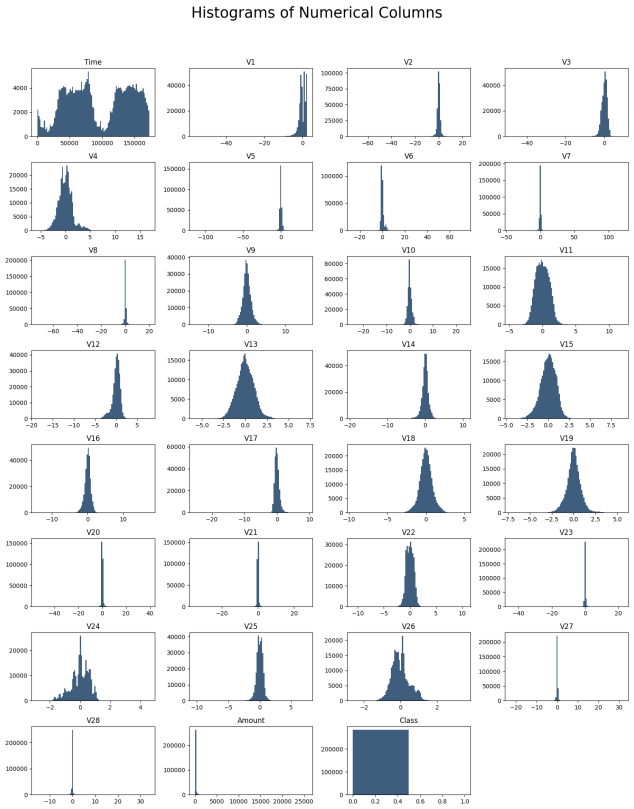



Figure 1. histograms of all features of dataset

We preprocessed the data using standard scalar technique and Then we use SMOTE oversampling technique to balance the imbalanced data and then we split the data as train and test data for train and test the machine and deep learning models.

## F.1. Logistic Regression:

The first model we implement is logistic regression we used a test size of 0.3 means that 30% of your data will be randomly selected and set aside for testing the model's performance after it has been trained. The remaining 70% of your data will be used to train the model.

These are the test results that has both performance metrics and heat map of confusion matrix of logistic regression.

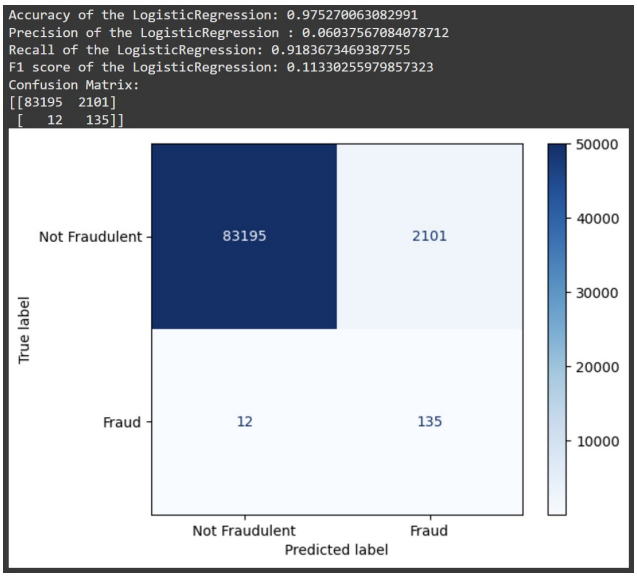
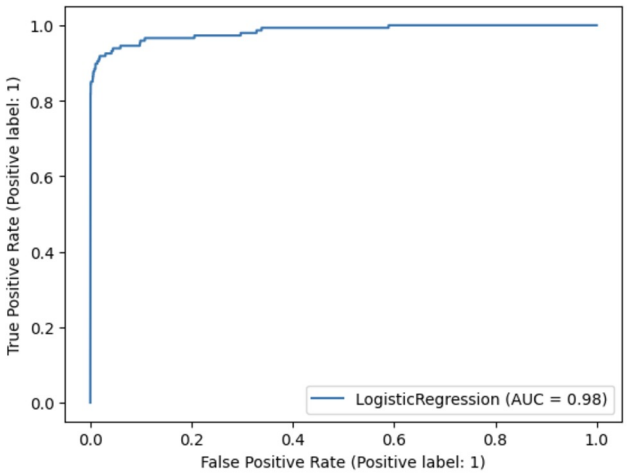



Figure 2. Test results of logistic regression

### F.1.1 ROC-AUC

The Receiver Operating Characteristics - Area Under the Curve (ROC-AUC) curve of Logistic Regression.




### F.1.2 Linear Regression on whole dataset:

The heat map and confusion matrix of logistic regression on whole dataset to find total number of fraudulent transactions.

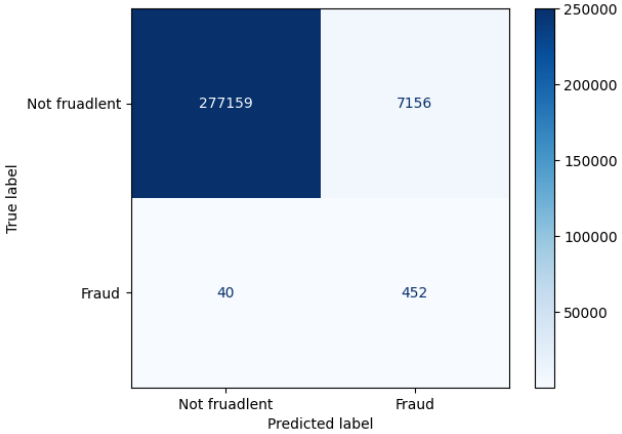



Figure 3. heat map of LR on whole dataset

## F.2. Random Forest Classifier:

The second machine algorithm we used is Random forest classifier we used an estimators of 100 for faster execution as random forest takes time.

These are the test results that has both performance metrics and heat map of confusion matrix of random forest classifier.

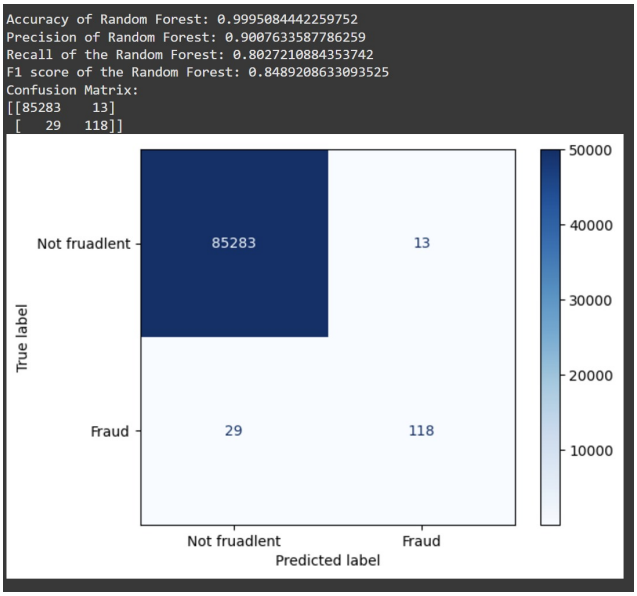
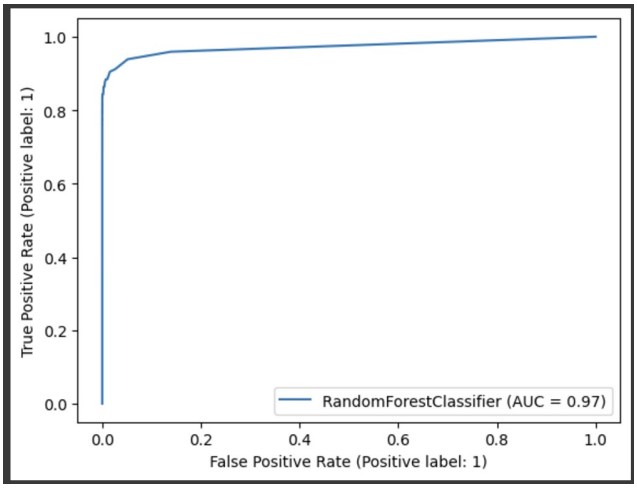



Figure 4. Test results of random forest

#### F.2.1 ROC-AUC:

The Receiver Operating Characteristics - Area Under the Curve (ROC-AUC) curve of Random Forest classifier




#### F.2.2 Random Forest on whole dataset

The heat map and confusion matrix of Random Forest on whole dataset to find total number of fraudulent transactions.

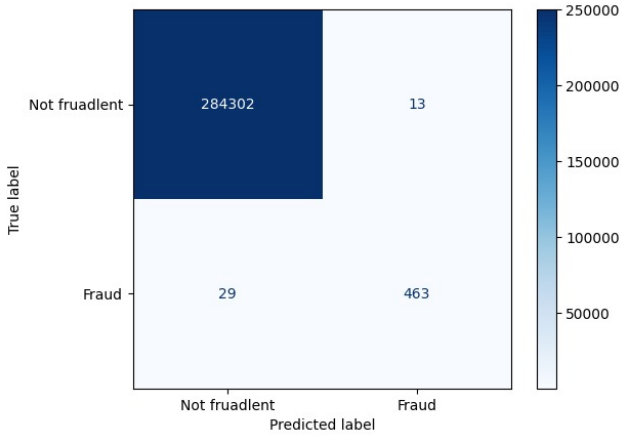



Figure 5. Enter Caption

### F.3. Decision Tree Classifier:

The third algorithm we implemented is decision tree classifier with critireon as entropy and random state 0.

These are the test results that has both performance metrics and heat map of confusion matrix of Decision tree classifier.

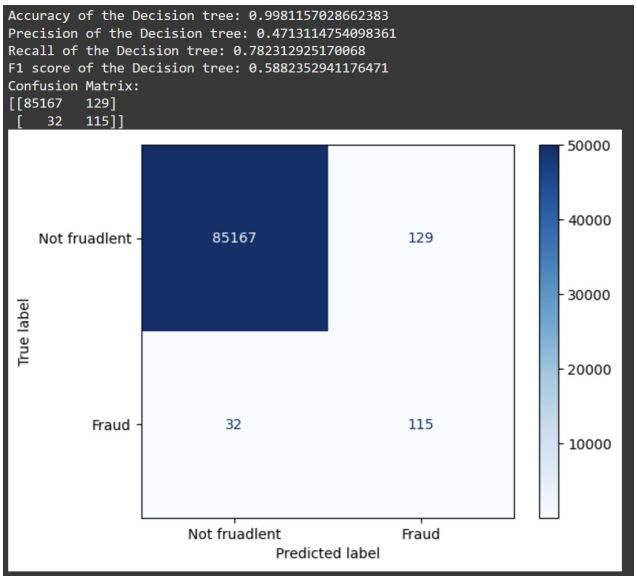
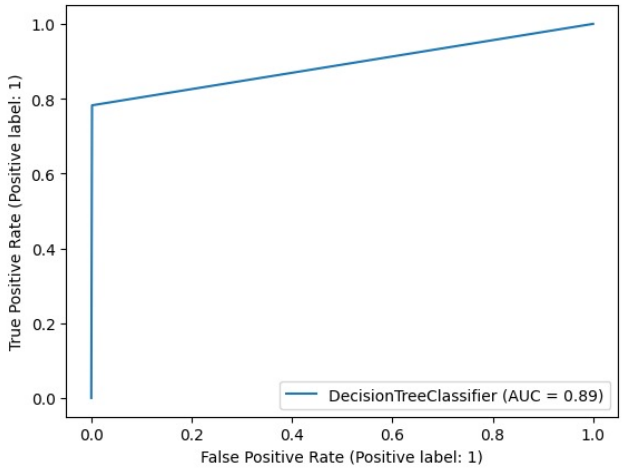



Figure 6. Test results of decision trees

#### F.3.1 ROC-AUC

The Receiver Operating Characteristics - Area Under the Curve (ROC-AUC) curve of Decision Tree classifier




#### F.3.2 Decision Tree on whole dataset

The heat map and confusion matrix of Decision Tree Classifier on whole dataset to find total number of fraudulent transactions.

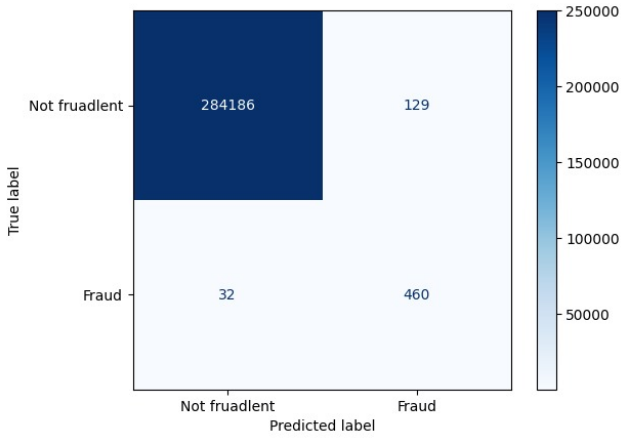



Figure 7. Decision tree on whole dataset

## F.4. Deep Neural Networks:

The final model we implemented is a deep learning algorithm. In this we used deep neural networks with five dense layers and one dropout layer and we used ”adam” as the optimizer.

These are the test results that has both performance metrics and heat map of confusion matrix of Deep neural networks.

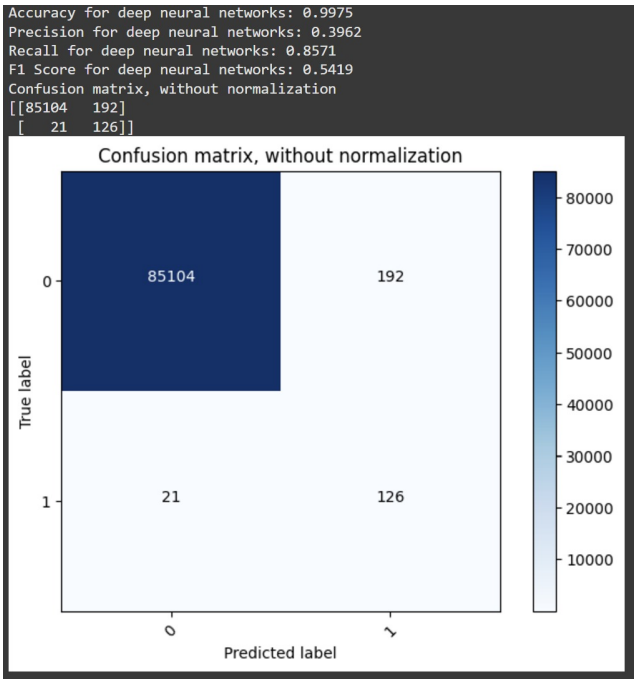
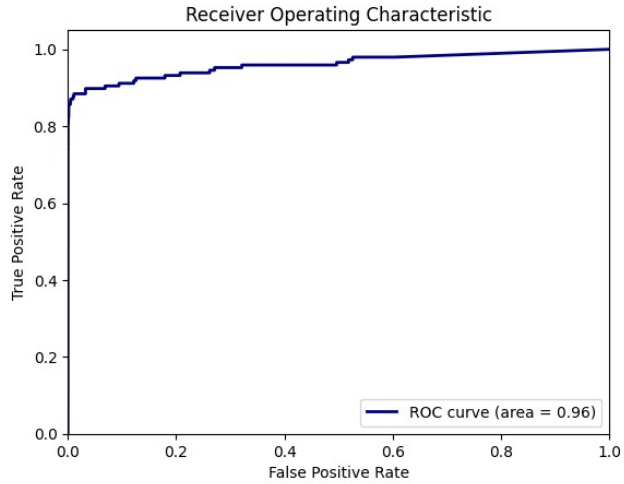



Figure 8. Test results of deep neural networks

### F.4.1 ROC-AUC

The Receiver Operating Characteristics - Area Under the Curve (ROC-AUC) curve of Deep Neural Networks




### F.4.2 Deep Neural Networks on whole dataset

The heat map and confusion matrix of Deep Neural Networks on whole dataset to find total number of fraudulent transactions.

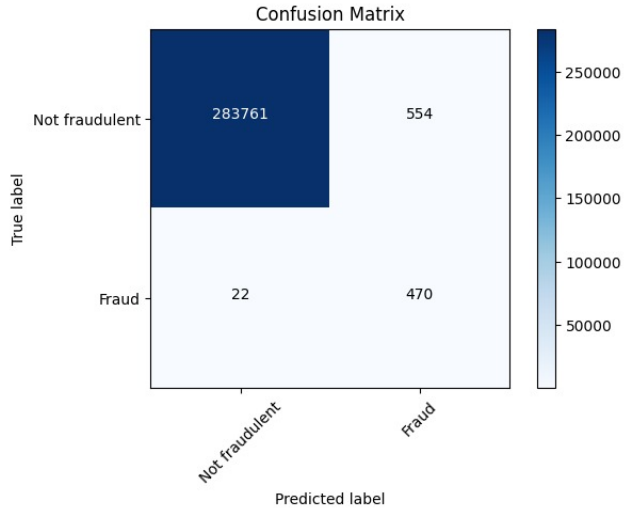



Figure 9. DNN on whole dataset

These are the results of Logistic regression, Random Forest, Decision trees and Deep neural networks. These also have performance metrics such as Accuracy, Precision, Recall and F1 score along with these they also have ROC-AUC and also heat map of confusion matrix of each algorithm implemented.

# G. CONCLUSION

Among all the given models here ,logistic regression turns out to be the worst as it marked 7156 non fraudulent transactions as fraudulent .

The logistic regression has taken the least time to evaluate the results among all the training models .

The random forest is the best model as it found 463 fraudulent transactions among 492 and it also only marked 13 non fraudulent transactions as fraudulent.

The random forest also takes the maximum time compared to others .

The decision tree found the fraud of 460 out of 492 transactions and it marked 129 non fraudulent as fraud .

Even though Deep Neural networks found most of the fraudulent transactions among all the models which is 470 out of 492 but it also marked 554 non Fraudulent as Fraudulent .

This model can be improved further by changing the optimizers and by adding more layers to it .

Results show that while traditional ML models like random forests and decision trees show strong performance, DNNs achieve higher accuracy and fewer false positives, making them better suited for complex fraud detection scenarios.

In conclusion, DNNs outperform traditional ML models by effectively identifying intricate fraud patterns. Future work could explore advanced architectures, such as autoencoders, and techniques to address data imbalance for further improvements in fraud detection systems.

# H. DIVISION OF LABOR

## H.1. TASKS BY ABHIPSHA SAHOO :

**1. Data Collection and Preprocessing :** Identified and sourced the dataset from Kaggle, ensuring it included features relevant for fraud detection.

Performed data cleaning tasks in the dataset.

Removed all null and duplicate values.

Handled outliers using statistical methods.

Standardized and normalized numerical features to improve model performance.

The Addressed class is imbalanced using Synthetic Minority Oversampling Technique (SMOTE).

**2.Implementation of Traditional Machine Learning Models**

### H.1.1 Logistic Regression:

Designed and implemented logistic regression for baseline performance measurement. Assessed model performance using metrics like accuracy, precision, recall and the f1 score , ROC AUC and Confusion matrix .

### H.1.2 Random forest :

Created a random forest ensemble model, exploring different numbers of trees and split criteria. Tuned hyper parameters e.g., max depth, number of estimators . for better and fast results.

**Report Contributions** Data pre processing methods And Implementation of details and results for Logistic Regression, Decision Trees, and Random Forest.

**Presentation Contributions** Created the slides for the Data pre processing techniques and class imbalance handling. Explanation of Logistic Regression , and Random Forest models. Comparative analysis of ML models.

## H.2. TASKS BY YESHASWEE VOLETY

### H.2.1 Implementation of Deep Neural Networks (DNNs)

Designed the architecture for a deep neural network: Input, hidden layers (with ReLU activation), and output (using sigmoid activation for binary classification).

Configured dropout layers to prevent over fitting. Preprocessed data for compatibility with the DNN model Split data into training, validation, and test sets and Converted data into tensors for model training. Trained the DNN model UsIing the Adam optimizer and binary cross-entropy loss function. Tuned hyperparameters like learning rate, batch size, and the number of hidden units for optimal performance.

### H.2.2 Decision Trees:

Developed and optimized a decision tree model with various depth constraints to prevent over fitting.

Analyzed feature importance from the model to interpret the decision-making process.

### H.2.3 report contribution

DNN implementation and performance analysis sections. Discussion on the superiority of DNN in handling complex fraud detection cases.

### H.2.4 Designed slides for:

DNN architecture, training process, and performance evaluation. Comparative metrics of traditional ML models versus the DNN. Future work and recommendations.

# I. REFERENCES

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